Problem A. Greedy arrays

Input file: `standard input`
Output file: `standard output`

You are given an array $a[1..n]$ of positive integers and an integer `type`.

For a target sum S, define the following greedy algorithm:

1. Sort all elements of the array in non-increasing order.
2. Traverse the elements in this order. Maintain the current remainder r, initially $r := S$.
3. For each current element x, if $x \le r$, take x and update $r := r - x$. Then move to the next element.
4. The algorithm is successful if, after processing all elements, the remainder becomes $r = 0$ (i.e. it selects a subset whose sum is exactly S).

Call an array greedy if for every sum S that can be obtained as the sum of some subset of its elements, the greedy algorithm described above is always successful in forming exactly S.

You are allowed to add exactly one positive integer t to the array, producing $a \cup \{t\}$. Positive integer t is valid if after adding t the resulting array is greedy.

Your task depends on the value of `type`:

- If `type` $= 0$, determine whether the number of valid values of t is finite or infinite.
- If `type` $= 1$, output all integers $t > 0$ such that after adding t the resulting array is greedy, or determine that there are infinitely many such t.

Input

The first line contains two integers n and `type` ($1 \le n \le 10^5$, `type` $\in \{0, 1\}$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^5$).

Output

- If `type` $= 0$, print exactly one line:
  - `finite` if the number of valid values of t is finite, otherwise print `infinite`.
- If `type` $= 1$, print:
  - on the first line print `finite` if the number of valid values of t is finite, otherwise print `infinite`.
  - if the answer is `finite`, then:
    - on the second line print an integer k — the number of valid values of t;
    - on the third line print k integers — all valid t in **increasing** order.

Scoring

This task contains 6 subtasks.

Subtask	Additional restrictions	Points
0	Examples	0
1	$n \le 10, a_i \le 40$	9
2	`type`$= 0$; $n \le 100, \sum a_i \le 10^5$	15
3	`type`$= 0$	20
4	$n,\ a_i \le 10^4$	20
5	$a_i \le 5 \cdot 10^4$	20
6	No additional restrictions	16

Examples

standard input	standard output
3 1 1 2 4	infinite
5 1 1 1 4 4 5	finite 3 1 2 3
4 0 2 2 3 5	finite
6 1 30 2 12 5 1 24	finite 3 3 4 6